Partial Differential Equations (PDE)
Assignment 2

[Wave equation]

(a) To show the derivative of integral: $\frac{d}{dt}\int_{a(t)}^{b(t)} f(x)dx = f(b(t))b'(t) - f(a(t))a'(t)$.

(b) Prove the solution to the problem $u_{tt} - c^2 u_{xx} = 0$ is $u(x,t) = f(x+ct) + g(x-ct)$, using (1) the variable $v = u_t + cu_x$ and (2) the coordinate method, $\xi = x + ct$ and $\eta = x - ct$, (3) and confirm your answer by direct substitution

(c) Prove the solution to the initial value problem $u_{tt} - c^2 u_{xx} = 0$ with the initial conditions $u(x,0) = \phi(x)$ and $u_t(x,0) = \psi(x)$ is

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds,$$

and confirm your answer by direct substitution

(d) Compute the solution of the initial value problem $u_{tt} - c^2 u_{xx} = 0$ with the initial conditions $u(x,0) = \log(1+x^2)$ and $u_t(x,0) = 4 + x$.